

# A Structure-Agnostic SHM Decision Framework for Reusable CFRP Spacecraft Structures

from Real-Data Detection to Physics-Grounded Life Prognosis

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Keio University   /   Composites Part B (draft)

# The reusable-vehicle decision

Reusable launch vehicles change the SHM question:

- not “did it survive one flight?”
- but **may it fly again — and for how many flights?**

CFRP primary structures (interstage, fairing, motor case, tank, drive shaft) fail by *internal* delamination / debond — invisible, inferred from NDT.

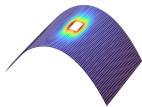
**Gap:** no end-to-end pipeline that propagates uncertainty from detection to a re-flight decision *and* states what real data supports.

detect → classify → characterise  
→ prognose → clear  
→ fleet-learn → design

# One decision core, five structures, four load modes

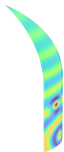
A structure-agnostic SHM decision framework across five reusable CFRP spacecraft structures and four load modes

Perforated interstage



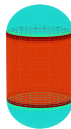
load: in-plane / membrane  
sensing: surface-stress GNN

H3 fairing



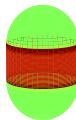
load: transverse press. / bending  
sensing: guided-wave GNN

SRB-3 motor case



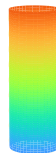
load: internal-pressure burst  
sensing: acoustic emission

CFRP COPV tank



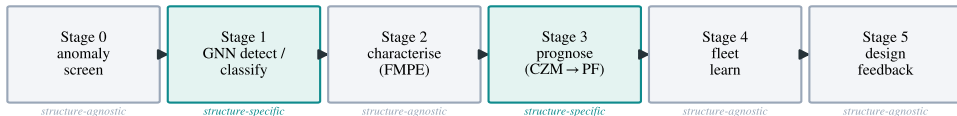
load: internal pressure  
sensing: shared  $|z|$  screen

CFRP drive shaft



load: TORSION (shear)  
sensing: shared  $|z|$  screen

Shared structure-agnostic decision core --- only detection (Stage 1) & prognosis physics (Stage 3) are structure-specific (pluggable)



detect → classify → characterise → prognosis → clear (OK/REPAIR/RETIRE) → fleet-learn → design | 1,499 real-FEM states, one margin surrogate  $R^2=0.94$

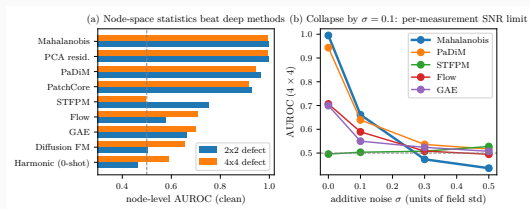
- Shared, structure-agnostic decision core (Stage 0/2/4/5 + clearance).
- Only *detection* (Stage 1) and *prognosis physics* (Stage 3) are structure-specific — pluggable.

# The five structures span the load-mode space

| structure          | load mode       | detection front-end   | evidence        |
|--------------------|-----------------|-----------------------|-----------------|
| interstage         | membrane        | surface-stress GNN    | FEM + meas. TSA |
| fairing            | transv./bending | guided-wave GNN       | FEM + real OGW  |
| SRB-3 case         | pressure burst  | acoustic emission     | FEM + real AE   |
| <b>drive shaft</b> | <b>torsion</b>  | shared $ z $ (future) | CZM-FEM         |
| COPV tank          | int. pressure   | shared $ z $ (future) | FEM             |

- Three demonstrated end-to-end with **real data**; two extend the load-mode coverage with FEM prognosis.

# Detection front-ends — grounded in real data



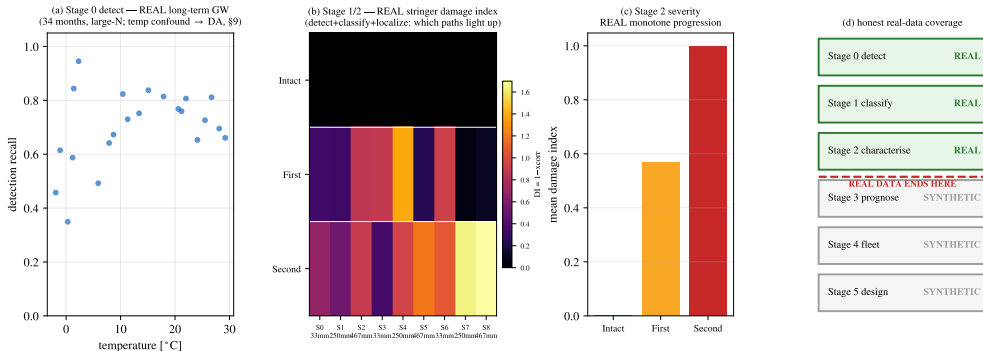
Stage-0  $|z|$  screen: per-node statistic dominates (AUROC  $\approx 0.99$ , fixed geometry).

## Structure-specific Stage-1 detectors:

- interstage: surface-stress GNN, macro-F1 0.76 (vs GAT 0.61); detection generalises, localisation collapses OOD.
- fairing: guided-wave GNN, F1 0.86.
- SRB-3: real AE, per-hit AUROC 0.89 / per-specimen 1.00.

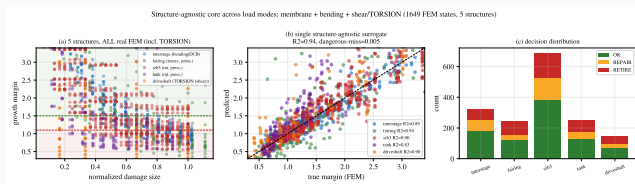
# How far does real data carry the pipeline? (honest)

Fairing (CFRP stringer) end-to-end on REAL Open-Guided-Waves data: detect → classify → characterise are real; prognosis is honestly synthetic



- Detect / classify / characterise on the fairing are **real data** (long-term guided waves, impact wavefield).
- **synthetic** from Stage-3 prognosis onward (representative phase-field, synthetic fleet) — the life side is the largest open gap.

# Structure-agnostic core — and its boundary

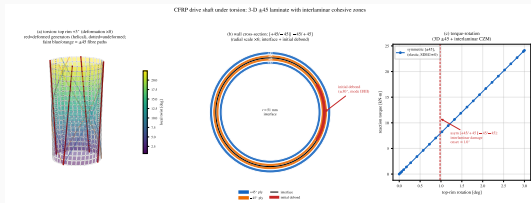


A single margin surrogate over **1,499 real-FEM** critical-load states fits all five ( $R^2=0.94$ , dangerous-miss 1.1%).

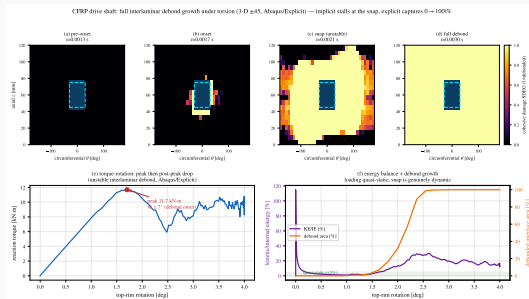
**Honest boundary:** leave-one-load-mode-out to the unseen torsion shaft *fails* ( $R^2 < 0$ ).

**Geometry transfers; load mode does not.**

## New load mode: the CFRP torsion drive shaft



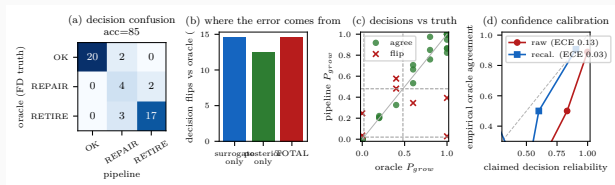
CGAX screen:  $\pm 45$  tube, groove damage,  $T_{cr}$  sweep.



3-D  $\pm 45^\circ$  + interlaminar CZM: full 0  $\rightarrow$  100% debond.  
Implicit stalls at the snap; **explicit** captures it.



# End-to-end decision reliability (vs a **synthetic** oracle)

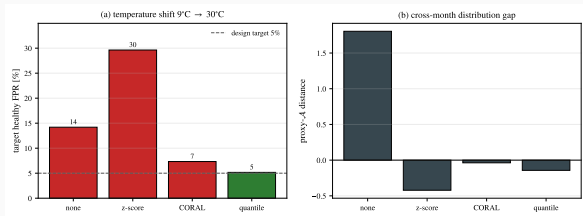


Propagate uncertainty to the clearance call; score against the exact FD-physics oracle.

- accuracy **85%**, dangerous-miss **0%**
- Platt recalibration: ECE  $0.18 \rightarrow 0.06$

Measured against a **synthetic oracle** — internal error propagation, not agreement with real life tests.

# Sim-to-real: normalisation choice controls false alarms

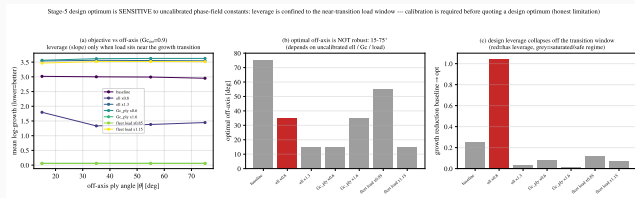


Real OGW temperature shift  $9^{\circ} \rightarrow 30^{\circ}\text{C}$ :

- uncorrected FPR **14%** (temperature mimics damage)
- quantile DA  $\rightarrow$  **5%** (design target)
- **z-score normalisation worsens to 30%**

Interstage TSA: quantile DA closes 97% of the gap.

# Stage-5 design feedback — mechanism, not a calibrated value



Fleet-driven inverse design steers off-axis angle + interface toughness.

**Honest limitation:** under uncalibrated phase-field constants the optimal angle spans 15°–75° — leverage exists only near the growth transition.

The **mechanism** is robust; a quoted optimum needs calibration.

- **real data**: detection / classification / characterisation (guided waves, AE, TSA), posterior coverage, 1,499-state FEM core.
- **synthetic**: prognosis, clearance, fleet, design — representative phase-field + synthetic fleets; decision UQ vs a synthetic oracle.
- **Largest gap**: no real life-test validation of the life side.
- Priority inputs: Kojima TSA full-field (interstage), real fatigue data.

- The differentiable, fast forward (FNO; autograd-exact) is what a TMCMC / Bayesian-inverse phase needs.
  - Its physics — anisotropic phase-field seeded by interlaminar cohesive zones — is the prognosis layer the torsion shaft extends from mode-I to mode-II/III shear.
- ⇒ **Phase-field** × **GNN prognosis**, calibrated on measured data, is the computational-solid-mechanics continuation.

One decision core across four load modes.

Real where data permits, honest where it does not.

Closing the life-side real-data gap is next.

Appendix / backup

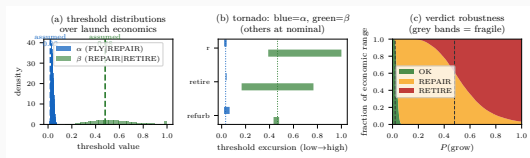
## A1 — Stage-0 anomaly benchmark (clean, fixed geometry)

| method               | 2×2   | 4×4   |
|----------------------|-------|-------|
| per-node Mahalanobis | 0.999 | 0.995 |
| PCA residual         | 0.999 | 0.995 |
| PaDiM                | 0.964 | 0.944 |
| PatchCore            | 0.927 | 0.918 |
| STFPM                | 0.754 | 0.496 |
| graph autoencoder    | 0.663 | 0.700 |
| flow (RealNVP)       | 0.578 | 0.707 |
| diffusion (FM)       | 0.503 | 0.663 |
| harmonicity (0-shot) | 0.465 | 0.590 |

node-level AUROC. Per-node statistics dominate on fixed geometry; the defect sits below  $0.1\sigma$  (a per-measurement SNR limit).



## A2 — Decision rule and launch economics



$a^* = \arg \min_a \mathbb{E}_{p(\theta|x)}[C(a, \theta)]$ , actions {OK, REPAIR, RETIRE}.

- MC over refurb/loss 1/100–1/30, retire/loss 0.1–0.4
- $\alpha = 0.035$  [0.021, 0.055],  $\beta = 0.53$  [0.28, 0.93]
- REPAIR-vs-RETIRE flips at  $P_{\text{grow}} \approx 0.3$ –0.5 (economics-fragile)

## A3 — Stage-2 characterisation (FMPE) vs differentiable physics

| set     | method    | $c_x$ RMSE   | size RMSE    | 90% cov.     |
|---------|-----------|--------------|--------------|--------------|
| in-dist | FMPE      | 0.006        | 0.002        | 0.958        |
| in-dist | diff-phys | 0.001        | 0.003        | 1.000        |
| OOD     | FMPE      | <b>0.144</b> | <b>0.096</b> | <b>0.000</b> |
| OOD     | diff-phys | 0.002        | 0.001        | 0.958        |

Forward physics generalises; amortised inference extrapolates poorly  $\Rightarrow$  hybrid Stage-2 (amortised in-dist, physics-grounded OOD).

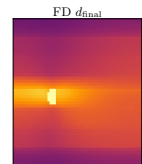
## A4 — Stage-3 prognosis: anisotropic phase-field

$$\Psi(\mathbf{u}, d) = \int_{\Omega} \left[ g(d) \psi^+(\boldsymbol{\varepsilon}) + \frac{G_c}{c_w} \left( \frac{d^2}{\ell} + \ell \nabla d \cdot \mathbf{A} \nabla d \right) \right] d\Omega$$

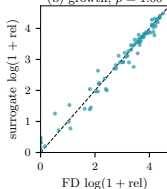
- $\mathbf{A} = \mathbf{I} + \beta \mathbf{a} \otimes \mathbf{a}$ : fibre-aligned crack resistance; interface  $G_c^{\text{int}}/G_c^{\text{ply}} \approx 0.3\text{--}0.6$  (DCB/ENF anchored).
- Carrara-2020 fatigue history degrades  $G_c$  across flights (no i.i.d. shortcut).
- Pluggable: fairing skin-core debond ( $G = \kappa \sigma^2 a^2$ , Paris); SRB-3 burst; *torsion* interlaminar CZM.

# A5 — Fast surrogate + trust gate

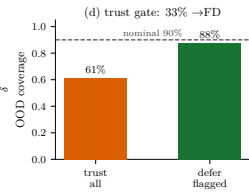
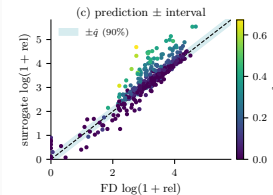
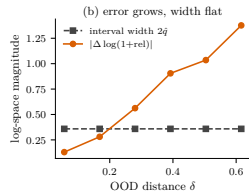
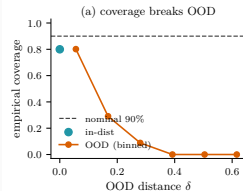
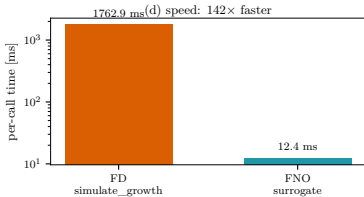
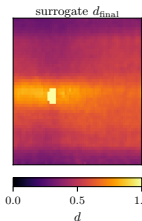
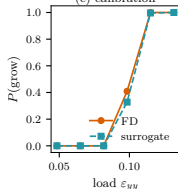
(a) load = 0.140



(b) growth,  $\rho = 1.00$



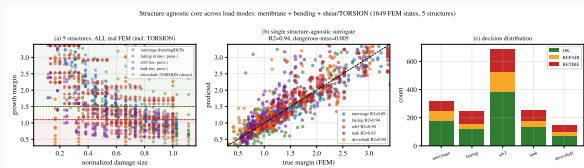
(c) calibration



Split-conformal trust gate: flags 33% of an OOD stream, restores coverage 61%  $\rightarrow$  88%.

FNO: field  $\text{rel-}L_2$  3.5%, grow-flag 0.988, **142x** speed.

# A6 — Structure-agnostic surrogate: details



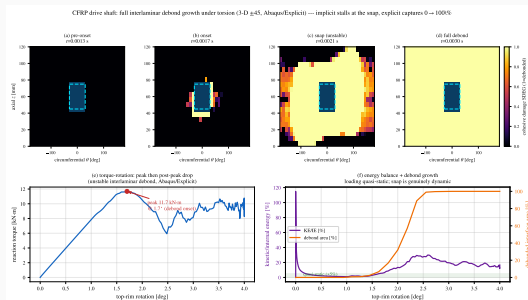
1,499 real-FEM states; descriptors [dmg, crit, aux, stiffness] + structure one-hot.

- per-structure  $R^2$ : interstage 0.91, fairing 0.94, SRB-3 0.93, tank 0.71, drive shaft 0.91
- shared rule: acc 0.86, dangerous-miss 1.1%
- LOSO torsion:  $R^2 = -1.4$ , dangerous-miss 17% (load-mode boundary)

# A7 — Torsion shaft: CGAX screen + 3-D explicit CZM

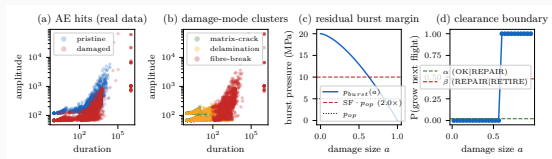
## CGAX screen ( $\pm 45^\circ$ tube, twist DOF):

- analytic check: wall shear  $17.1$  vs  $T/2\pi R^2 t = 16.8$  MPa
- 144-state groove sweep,  $T_{cr}$   $0.31$ – $7.76$  kN·m
- true torsional buckling is non-axisymmetric  $\Rightarrow$  3-D



3-D explicit: KE/IE  $1.3$ – $3.6\%$  in loading (quasi-static), peak  $11.7$  kN·m @  $1.7^\circ$ , debond  $0 \rightarrow 100\%$ .

## A8 — SRB-3 acoustic-emission detection (real data)

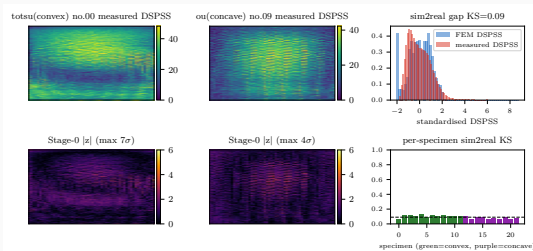


4TU CC0 AE,  $\sim 48k$  Vallen hits:

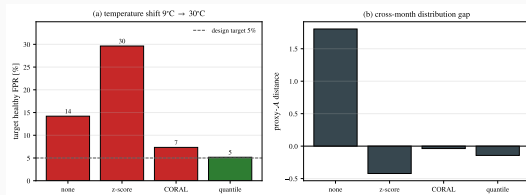
- group-aware RF: per-hit AUROC 0.888 / per-specimen 1.000 (linear baseline  $\sim 0.52$ )
- amplitude–duration damage-mode clustering

**Honest:** compression-pristine coupons also emit high-energy AE  $\Rightarrow$  the multivariate detector, not one cluster, is the read-out; AE is a real-data *proxy* for the motor-case front-end.

# A9 — Sim-to-real domain adaptation: details



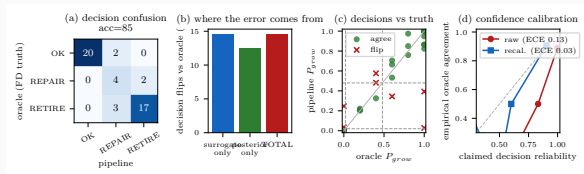
Interstage TSA: proxy- $\mathcal{A}$  2.0  $\rightarrow$  0.3 (97% closed).



Fairing OGW: FPR 14  $\rightarrow$  5% (quantile), CORAL 7%, z-score  $\rightarrow$  30%; threshold set on source healthy (no target leak).



# A10 — End-to-end decision UQ: error decomposition

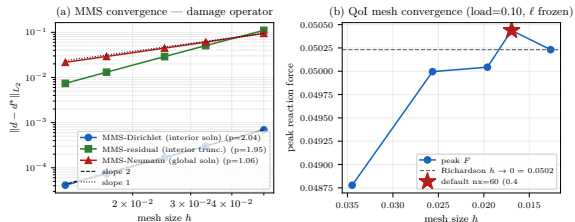


Four decisions isolate each error (over 40 held-out scenarios):

- Stage-2 posterior: flip rate 25% (dominant)
- surrogate forward: 17.5%
- partial cancellation → total 15%
- all 16 RETIRE cases caught (0% dangerous-miss)

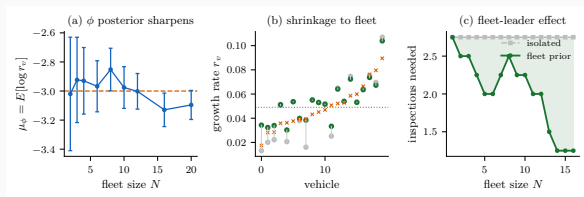
Oracle = exact FD physics (**synthetic**).

# A11 — Solver verification and robustness



- MMS: interior order 2.04 (2-D), 2.05 (3-D)
- mesh convergence: 0.4% at  $n_x=60$
- constant-peak proxy under-predicts life  $3.4\times$  vs a variable flight spectrum (S-N anchored)
- Stage-0 holds AUROC 0.84 at  $\sigma=0.1$ ; chance near  $\sigma \approx 0.7-1.0$

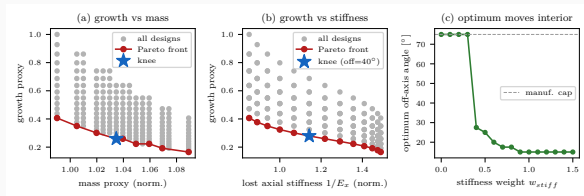
# A12 — Stage-4 hierarchical fleet learning



Conjugate-Gibbs partial pooling over a synthetic fleet ( $N=20$ ,  $F=30$ ):

- recovers  $\varphi$ :  $\mathbb{E}[\mu] = -3.10$  (true  $-3.0$ ),  
 $\mathbb{E}[\sigma] = 0.44$  (true  $0.45$ )
- prior SD sharpens  $0.39 \rightarrow 0.10$
- fleet-leader: stable clearance in 1.25 vs 2.75 inspections

# A13 — Stage-5 multi-objective design (Pareto)

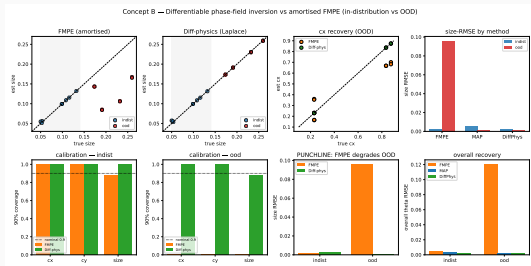


Growth vs. mass / axial-stiffness ( $A_{11}$ ) / bending ( $D_{11}$ ):

- single growth objective rails off-axis to the manufacturable cap
- competing CLT objectives restore an interior optimum (knee  $\sim 40^\circ$ )
- optimal angle  $75^\circ \rightarrow 15^\circ$  as stiffness weight grows

(sensitivity to uncalibrated constants: main deck.)

# A14 — Differentiable inversion (Keio bridge)



JAX AT2 forward, FD-verified gradient ( $3 \times 10^{-7}$ );  
Laplace posterior  $\Sigma = \sigma^2(J^\top J + \Lambda)^{-1}$ , no labelled  
pairs.

- amortised collapses OOD ( $\sim 25\times$ , 0 coverage)
  - differentiable-physics invariant
  - FNO-as-forward: MAP+Laplace 530s  $\rightarrow$  12s ( $43\times$ )
- $\Rightarrow$  the cheap forward TCMCMC needs (Keio Phase-3).

## A15 — Data provenance and honesty

- **real data**: interstage TSA full-field (Kojima, CPB 2025 — **co-author / consent gated**); fairing long-term + impact guided waves (OGW); SRB-3 AE (4TU CC0); 4TU DCB fatigue.
- **synthetic**: phase-field forward (representative  $G_c, \ell, \beta$ , fatigue); synthetic fleet; decision-UQ oracle = exact FD physics.
- Reproducibility:  $\sim 250$  unit tests, one-command end-to-end run, MMS / mesh / SNR / economic-sensitivity studies; figures script-generated.
- **Open gap**: no real life-test of the prognosis  $\rightarrow$  clearance  $\rightarrow$  fleet chain.